



EduBot – AI Study Assistant

AI Capstone Project SDG 4: Quality Education

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1. Selected SDG and Reason for Selection

Selected SDG: SDG 4 – Quality Education

The United Nations Sustainable Development Goal 4 focuses on ensuring inclusive and equitable quality education and promoting lifelong learning opportunities for all. It aims to provide free, equitable, and quality primary and secondary education to every child by 2030.

Why This SDG?

Education is the foundation of all development. However, millions of students — especially in developing countries like India — lack access to:

- Personalized tutoring and academic support
- 24/7 learning assistance outside school hours
- Affordable and accessible study resources
- Guidance on career paths and subject choices

EduBot directly addresses these gaps by providing AI-powered, free, and always-available study assistance to any student with internet access.

2. Problem Statement

In India and across the world, a large percentage of students struggle with their studies due to lack of personalized support. Key problems include:

- Limited access to quality tutors, especially in rural and semi-urban areas

- High cost of private coaching and tutoring services
- No 24/7 support system available for students outside school hours
- Students feel hesitant to ask questions in class due to fear of judgment
- Lack of guidance on career paths and future opportunities

If this problem is not solved, millions of students will continue to fall behind academically, leading to higher dropout rates, unemployment, and widening inequality in education quality.

3. Proposed Solution

What is EduBot?

EduBot is an AI-powered chatbot built using Botpress that acts as a personal study assistant for students. It is available 24/7, completely free, and accessible through any web browser.

How It Works

EduBot uses a Generative AI Autonomous Node (powered by a Large Language Model) to understand student queries and provide intelligent, personalized responses. The bot guides students through a structured flow:

- Student opens EduBot and receives a welcome message
- Student chooses from 4 options: Learn a Topic, Quiz Me, Study Tips, or Career Guidance
- Based on the selection, the bot asks follow-up questions
- The AI Autonomous Node generates personalized, accurate responses
- Student receives explanations, quizzes, tips, or career suggestions

Who Will Use EduBot?

- School students (Class 6–12)
- College students seeking subject help
- Self-learners and competitive exam aspirants

4. Project Features

Feature	Description
 Learn a Topic	Student selects a subject (Math, Science, English, History, CS) and types any topic. AI explains it with a simple definition, real-life example, and a practice question.

 Quiz Me	Student selects a subject and gets 3 MCQ questions one by one. AI gives feedback after each answer and shows the final score.
 Study Tips	Student selects their level (School/College/Self Learner) and receives 5 personalized study tips tailored to their needs.
 Career Guidance	Student selects their area of interest and receives 2–3 career path suggestions with required skills and how to get started.
 AI Autonomous Node	Powered by a Large Language Model (LLM) through Botpress, enabling dynamic, context-aware responses to any student query.

5. Technology Stack

Technology	Usage
Botpress	Chatbot platform for building the conversational flow
Autonomous Node (LLM)	Generative AI engine that powers intelligent responses
Natural Language Processing	Understands and processes student queries in natural language
Web Chat Interface	Front-end interface for students to interact with EduBot
GitHub	Open source repository for project documentation and screenshots

6. Project Screenshots

The following screenshots demonstrate EduBot working across all major features:

- Screenshot 1: EduBot welcome screen and main menu with 4 options
- Screenshot 2: Learn a Topic flow — subject selection and AI explanation of Algebra
- Screenshot 3: Quiz Me flow — subject selection and quiz questions
- Screenshot 4: Study Tips flow — level selection and personalized tips
- Screenshot 5: Career Guidance flow — interest selection and career suggestions
- Screenshot 6: Full Botpress canvas showing complete flow diagram

All screenshots are available in the GitHub repository under the screenshots folder.

7. Open Source Repository

The complete project including flow documentation, screenshots, and README is available on GitHub:

GitHub Repository: <https://github.com/Anushree338/EduBot-SDG4>

Live Bot Demo: <https://anushree338.github.io/EduBot-SDG4>

The repository contains:

- README.md with complete project description
- 6 screenshots demonstrating all features
- Live bot link for direct testing

8. Future Scope

EduBot can be significantly enhanced in the future with the following improvements:

- Voice Support: Add voice input/output so students can speak their questions instead of typing
- Regional Language Support: Add Hindi, Marathi, Tamil, and other Indian languages to reach more students
- Progress Tracking: Track student performance over time and show improvement graphs
- Mobile Application: Build a dedicated Android/iOS app for easier access on smartphones
- School Portal Integration: Integrate EduBot directly into school learning management systems
- Personalized Learning Paths: Use AI to create custom study plans based on student weak areas
- Doubt Solving with Images: Allow students to upload photos of questions for AI to solve
- Competitive Exam Preparation: Add specialized modules for JEE, NEET, UPSC preparation

Conclusion

EduBot is a practical and impactful AI solution that directly addresses SDG 4: Quality Education. By leveraging Generative AI through Botpress, it provides personalized, 24/7 study assistance to students who cannot afford private tutoring. The bot is easy to use, covers multiple subjects, and guides students through learning, quizzing, study tips, and career guidance — all in one place.

This project demonstrates the power of Generative AI and conversational interfaces in solving real-world education challenges and making quality learning accessible to every student.

